

Oracle Database 12c: Datafile Restore Without Backup

Comprehensive Step-by-Step Technical Guide for Missing Datafile Recovery in RMAN

1. Overview & Initial Full Backup

Before performing recovery scenarios, establish a solid baseline by taking a complete full database backup including archived redo logs using Recovery Manager (RMAN).

ENVIRONMENT SETUP & SQL*PLUS STARTUP

```
[oracle@localhost Desktop]$ . oraenv
ORACLE_SID = [oracle] ? apple
SQL> startup
Database mounted.
Database opened.
```

RMAN FULL BACKUP COMMAND

```
[oracle@localhost Desktop]$ rman target /
RMAN> backup database plus archivelog;
```

Backup Summary Check: RMAN records full datafile backup sets ('TAG20210717T093353') and archived log backup sets, ensuring an anchor point for database roll-forward operations.

2. Creating Tablespace and User Objects

Next, we create a custom tablespace named **EXAMPLE** with an associated data file, followed by a dedicated schema user **LION** to hold our test data.

SQL EXECUTION

```
SQL> create tablespace example datafile '/u01/app/oracle/oradata/apple/example01.dbf' size 50m autoextend on;
SQL> create user lion identified by lion default tablespace example quota unlimited on example;
SQL> grant create session, create table to lion;
```

We then connect as user **LION**, create the **test** table, and populate it with rows:

```
SQL> create table test (id number(2));
SQL> insert into test values (34);
SQL> insert into test values (35);
SQL> insert into test values (45);
SQL> insert into test values (21);
commit;
```

3. Simulating Datafile Deletion Disaster

To simulate a critical production failure where a datafile is accidentally deleted from the OS level while the database is shut down or aborted:

ABORT INSTANCE & DELETE FILE

```
SQL> shu abort;
[oracle@localhost Desktop]$ rm /u01/app/oracle/oradata/apple/example01.dbf
```

Disaster Triggered: The physical datafile `example01.dbf` (Datafile ID 5) has been permanently removed from the storage disk outside of Oracle control.

4. Database Startup Failure

Attempting to restart the database instance results in an error because Oracle detects that Datafile 5 is missing during mount-to-open transition.

STARTUP OUTPUT

```
SQL> startup
ORACLE instance started.
Database mounted.
ORA-01157: cannot identify/lock data file 5 - see DBWR trace file
ORA-01110: data file 5: '/u01/app/oracle/oradata/apple/example01.dbf'
```

5. RMAN Restore and Media Recovery

With the database in mounted mode, RMAN can recreate the missing datafile pointer and restore the file contents, followed by media recovery to synchronize transactions.

RMAN RECOVERY COMMANDS

```
[oracle@localhost Desktop]$ rman target /
RMAN> restore datafile 5;
RMAN> recover database;
RMAN> alter database open;
```

RMAN Command	Action Description	Result
<code>restore datafile 5;</code>	Recreates the empty physical datafile on disk at the exact path.	Datafile header and blocks regenerated.
<code>recover database;</code>	Applies redo log changes up to the point of failure.	Media recovery complete.
<code>alter database open;</code>	Opens the database for read/write operations.	Database online and accessible.

6. Verification and Data Integrity Check

Finally, we log back into SQL*Plus to confirm all datafiles are online and query the user table `test` to verify that all inserted records are fully intact.

VERIFICATION QUERIES

```

SQL> select name, open_mode from v$database;
NAME          OPEN_MODE
-----
APPLE         READ WRITE

SQL> select name from v$datafile;
/u01/app/oracle/oradata/apple/system01.dbf
/u01/app/oracle/oradata/apple/sysaux01.dbf
/u01/app/oracle/oradata/apple/undotbs01.dbf
/u01/app/oracle/oradata/apple/users01.dbf
/u01/app/oracle/oradata/apple/example01.dbf

SQL> select * from lion.test;
      ID
-----
      34
      35
      45
      21

```

Conclusion: Datafile 5 was successfully restored and recovered without requiring a prior backup specifically for that tablespace, leveraging RMAN restore, archived redo logs, and database media recovery.